

Solution Architecture

Date	05 AUG 2026
Student Name	P. Srujan Sinha
Project Name	Intergalactic Space Travel Request (ISTR) System
Maximum Marks	4 Marks

1. Architecture Overview

ISTR is built as a scoped ServiceNow application layered across frontend, backend/logic, security, database, notification, and analytics tiers, all running on the ServiceNow SaaS cloud platform.

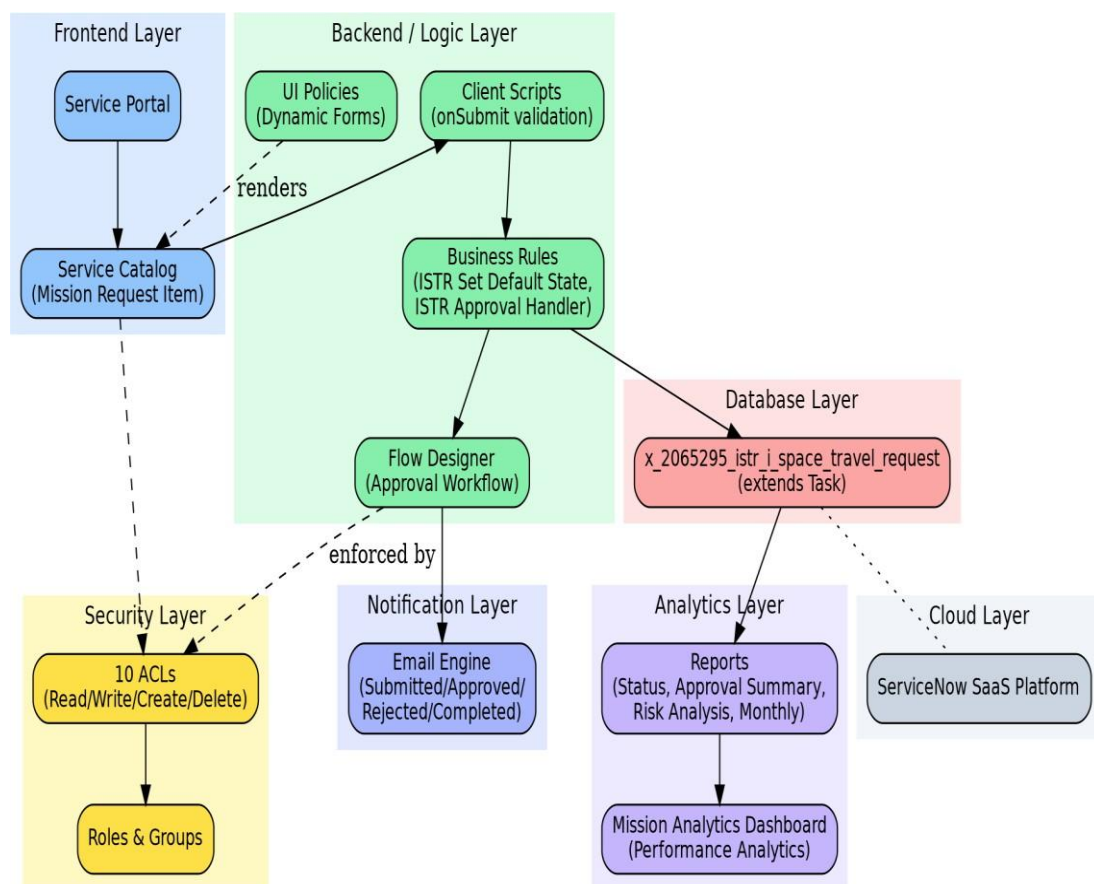


Figure 1: ISTR Solution Architecture

2. Technology Stack

Layer	Technology
Frontend	Service Portal, Service Catalog
Backend	Business Rules, Flow Designer, Client Scripts, UI Policies
Database	ServiceNow Custom Table (extends Task)
Security	ACLs, Users, Roles, Groups
Notifications	Email Engine

Layer	Technology
Reports	ServiceNow Reports
Dashboard	Performance Analytics
Cloud	ServiceNow SaaS

3. Database Design

3.1 Custom Table

Table name: x_2065295_istr_i_space_travel_request — extends Task, inheriting standard fields (number, state, short_description, assignment, etc.) plus the mission-specific fields below.

3.2 Key Fields

Field	Notes
Mission ID	Auto-numbered, inherited from Task
Mission Name	Free text
Mission Type	Choice: Research, Exploration, Colonization, Emergency Rescue, Cargo Transport, Diplomatic Mission, Alien Negotiation, Military Operation
Destination Planet	Free text / reference
Travel Date	Date/Time
Return Date	Date/Time
Risk Level	Choice: Low, Medium, High, Critical
Mission Description	Multi-line text
Crew Members	List / reference
Budget	Currency
Manager	Reference (sys_user)
Finance Officer	Reference (sys_user)
Mission Control Officer	Reference (sys_user)
Diplomatic Officer	Reference (sys_user) – shown conditionally
Mission Status	Choice: Draft, Submitted, Manager Approved, Finance Approved, Mission Control Approved, Rejected, Scheduled, Completed, Cancelled
Approval Status	Choice / rollup of approval state
Attachments	Standard attachment relationship

4. Security Model

4.1 Roles

- Mission Requester
- Manager

- Finance Officer
- Mission Control
- System Administrator

4.2 Groups

- Mission Requesters
- Managers
- Finance Team
- Mission Control Team
- System Administrators

4.3 ACLs

10 ACLs secure Create, Read, Write, and Delete operations across roles on the x_2065295_istr_i_space_travel_request table, ensuring each role can only access the operations and fields relevant to its responsibilities.

5. Automation Layer

5.1 Business Rules

Name	Purpose
ISTR Set Default State	Automatically sets the initial request state on creation
ISTR Approval Handler	Handles approval status updates as the request moves through workflow stages

5.2 Client Script

onSubmit script validates that when Mission Type = Alien Negotiation, a Diplomatic Officer must be specified before the form can be submitted.

5.3 UI Policy

When Mission Type = Alien Negotiation, the Diplomatic Officer field is shown; otherwise it is hidden.

6. Workflow (Flow Designer)

Mission Request → Manager Approval → Finance Approval → Mission Control Approval → Mission Scheduled → Completed → Closed.

7. Data Flow

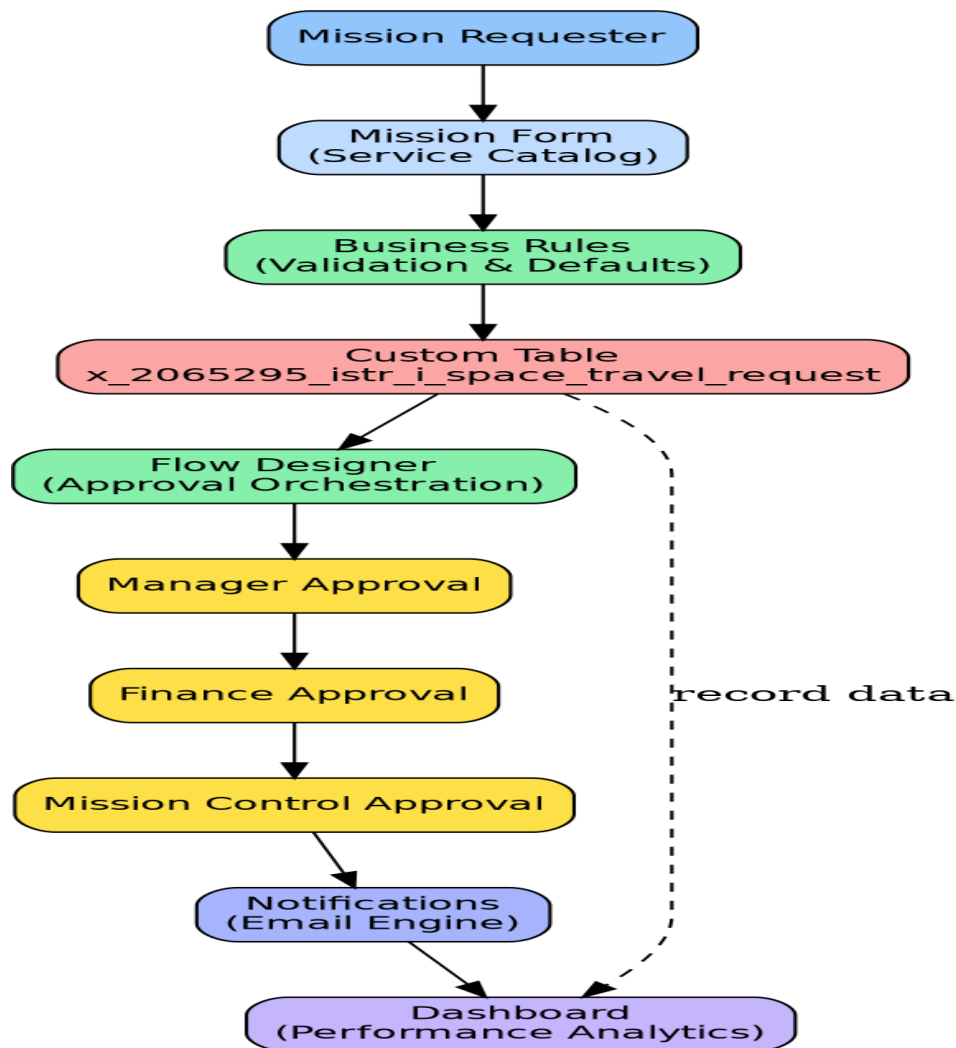


Figure 2: ISTR Data Flow – Mission Requester to Dashboard

8. Notifications

Trigger	Notification
Request Submitted	Confirmation email to requester
Request Approved (any level)	Status update email
Request Rejected	Rejection email with reason
Mission Completed	Completion email to requester and Mission Control

9. Analytics Layer

9.1 Dashboard – Mission Analytics

- Pending Requests
- Approved Requests
- Rejected Requests
- Mission Types
- Risk Levels
- Monthly Requests

9.2 Reports

- Mission Status
- Approval Summary
- Risk Analysis
- Monthly Requests

10. Summary

The ISTR solution architecture combines Service Portal/Catalog intake, Flow Designer-driven multi-level approval, Business Rule and Client Script validation, ACL-based role security, automated notifications, and Performance Analytics reporting into a single, cohesive ServiceNow scoped application built on a Task-extended custom table.